

Iterative Self-Refinement for Multi-Camera Calibration via Learned Pose Conditioning

Abstract

We present a method for improving multi-camera extrinsic calibration using iterative self-conditioning with a learned geometry model. Starting from Pi3X, a permutation-equivariant feed-forward network that estimates camera poses from images, we observe that feeding its own predictions back as conditioning improves rotation estimates but degrades translations due to Sim(3) scale ambiguity. We propose a hybrid fusion strategy that decouples rotation and translation channels: rotation edges are extracted from conditioned inference while translation edges are preserved from unconditioned inference. Combined with Monte Carlo noise perturbation, this achieves 28% rotation improvement and 21% relative pose improvement over the baseline, with no external calibration data required.

1 Introduction

We address the problem of extrinsic calibration for 8 static cameras deployed in a cattle feedlot. The scene presents challenges where classical calibration methods fail: textureless dynamic subjects (black cattle), minimal inter-camera overlap, and no physical calibration targets.

We use Pi3X [?], a permutation-equivariant feed-forward geometry model that processes all N views jointly and outputs camera poses up to Sim(3) ambiguity. By aggregating predictions across 91 timesteps, we obtain an initial calibration estimate with $\sim 4.4^\circ$ mean rotation error.

Key insight. Pi3X accepts optional pose conditioning—feeding known camera poses alongside images improves its predictions. We exploit this by feeding Pi3X’s *own output* back as conditioning, creating an iterative self-refinement loop. However, naïve iteration improves rotation but degrades translation due to Sim(3) scale ambiguity compounding across iterations.

Contribution. We propose a hybrid fusion strategy that:

1. Decouples rotation and translation in the self-refinement loop
2. Uses Monte Carlo noise perturbation to explore the posterior over poses
3. Achieves 28% rotation improvement with no external calibration data

2 Method

Our method consists of four stages, each building on the previous.

2.1 Stage 1: Baseline Pipeline

Given $K = 91$ timesteps of 8-camera images, we run Pi3X to obtain per-timestep poses $\{T_k^c\}_{k=1}^K$ for each camera $c \in \{1, \dots, 8\}$, where each $T_k^c \in \text{SE}(3)$ is defined up to a per-timestep Sim(3) gauge.

Gauge fixing. We align all timesteps to a reference frame via Sim(3) Procrustes (Umeyama’s method), removing the per-timestep scale and frame ambiguity.

Outlier rejection. For each camera, we project its K estimated positions onto the dominant 2D plane (via PCA) and reject outliers using the Median Absolute Deviation (MAD) with threshold $k = 3$.

Pose graph optimization. We extract pairwise relative poses from inlier timesteps, then optimize globally via:

- L_1 Weiszfeld rotation averaging on $SO(3)$
- Translation averaging with hard co-location constraints (cameras sharing a pole are forced to the same position)

2.2 Stage 2: Self-Conditioned Inference

Pi3X supports multimodal conditioning: given poses $\hat{T}^c$ and intrinsics K^c , it refines its predictions. We condition on the pipeline output from Stage 1:

$$\{T_k^{c,(1)}\} = \text{Pi3X}(\text{imgs}_k \mid \hat{T}^c, K^c), \quad k = 1, \dots, K \quad (1)$$

where $\hat{T}^c$ is the Stage 1 estimate converted to video camera order.

2.3 Stage 3: Hybrid Fusion

We observe that conditioned inference improves rotation but degrades translation (Section ??). To exploit both, we fuse edges from both sources in the pose graph:

$$\tilde{T}_{ij}^{\text{fused}} = \begin{pmatrix} R_{ij}^{\text{cond}} & t_{ij}^{\text{uncond}} \\ 0 & 1 \end{pmatrix} \quad (2)$$

where R_{ij}^{cond} is the mean relative rotation from conditioned inference and t_{ij}^{uncond} is the mean relative translation from unconditioned inference. Edge weights are taken as $w_{ij} = \max(w_{ij}^{\text{cond}}, w_{ij}^{\text{uncond}})$.

The initial pose for optimization uses conditioned rotations with unconditioned translations:

$$T_{\text{init}}^c = \begin{pmatrix} R_{\text{cond}}^c & t_{\text{uncond}}^c \\ 0 & 1 \end{pmatrix} \quad (3)$$

2.4 Stage 4: Monte Carlo Iterative Refinement

To further improve rotation estimates, we iterate the hybrid fusion with stochastic perturbation:

Key design choices:

- **Noise:** Gaussian perturbation with $\sigma_R = 2$ (axis-angle) and $\sigma_t = 0.05$ (isotropic)
- **Translation anchoring:** Translations are always sourced from the unconditioned baseline, never updated from conditioned outputs
- **Early stopping:** Monitor ARE convergence; terminate when improvement < 0.1

Algorithm 1 Iterated Hybrid MC Refinement

Require: Unconditioned pipeline output $\hat{T}^c$, images, intrinsics

Require: Noise parameters σ_R, σ_t ; samples S ; iterations M

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1:  $T_{\text{current}} \leftarrow \hat{T}^c$ 
2: Compute unconditioned translation edges  $\{t_{ij}^{\text{uncond}}, w_{ij}^t\}$ 
3: for  $m = 1$  to  $M$  do
4:   for  $s = 1$  to  $S$  do
5:      $T_s \leftarrow \text{AddNoise}(T_{\text{current}}, \sigma_R, \sigma_t)$ 
6:      $\{T_{k,s}^c\} \leftarrow \text{Pi3X}(\text{imgs} \mid T_s, K^c)$  for all timesteps
7:      $\hat{T}_s^c \leftarrow \text{Pipeline}(\{T_{k,s}^c\})$  {gauge fix + outlier rej}
8:   end for
9:   Extract rotation edges  $\{R_{ij}^{\text{cond}}, w_{ij}^R\}$  from all  $S \times K$  aligned poses
10:  Fuse:  $\tilde{T}_{ij} = (R_{ij}^{\text{cond}}, t_{ij}^{\text{uncond}})$ 
11:   $T_{\text{opt}} \leftarrow \text{PoseGraph}(\tilde{T}_{\text{init}}, \tilde{T}_{ij}, \max(w^R, w^t))$ 
12:   $T_{\text{current}} \leftarrow (R_{\text{opt}}, t_{\text{uncond}})$  {preserve translations}
13: end for
14: return  $T_{\text{opt}}$ 
```

3 Results

3.1 Experimental Setup

Dataset. 8 static cameras (2688×1520) in a cattle feedlot, 91 timesteps sampled at interval 10 from synchronized video. Ground truth poses from COLMAP.

Evaluation metrics.

- **ATE:** Absolute Translation Error (RMSE after Sim(3) alignment)
- **ARE:** Absolute Rotation Error (mean angular error in degrees)
- **RPE rot/trans:** Relative Pose Error for all camera pairs
- **C2C3, C5C6:** Distance between co-located camera pairs (GT ≈ 0)

3.2 Comprehensive Comparison

Table 1: Comprehensive comparison of all methods. ATE and RPE trans are in arbitrary (Sim(3)-aligned) units. Cost is relative to a single unconditioned Pi3X inference pass over 91 timesteps (~ 100 s on an A100).

Method	ATE↓	ARE↓	RPE rot↓	RPE trans↓	C2C3↓	Cost
1. Raw Pi3X (quat average)	0.177	4.37°	6.26°	0.395	0.088	1×
2. + Pipeline	0.161	4.39°	6.26°	0.375	0.000	1×
3. + Iterative Refinement	0.166	3.49°	4.93°	0.384	0.000	2×
4. + Hybrid Refinement	0.159	3.45°	4.93°	0.322	0.000	2×
5. + Hybrid MC (3×4)	0.160	3.13°	4.57°	0.311	0.000	13×

3.3 Summary of Improvements

Comparing the full method (row 5) against raw Pi3X (row 1):

- **ARE:** $4.37^\circ \rightarrow 3.13^\circ$ (**28% improvement**)
- **RPE rot:** $6.26^\circ \rightarrow 4.57^\circ$ (**27% improvement**)
- **RPE trans:** $0.395 \rightarrow 0.311$ (**21% improvement**)
- **ATE:** $0.177 \rightarrow 0.160$ (**10% improvement**)
- **Co-location:** $0.088 \rightarrow 0.000$ (hard constraint enforced)

The hybrid refinement (row 4) offers the best cost-benefit tradeoff: $2\times$ compute for 21% rotation and 18% RPE improvement.

4 Ablation Studies

4.1 Why Hybrid Fusion is Necessary

Naïve iterative refinement (feeding full conditioned output back without decoupling) degrades translation while improving rotation:

Table 2: Naïve vs. hybrid iterative refinement over 5 iterations.

Method	ATE↓	ARE↓	Δ ATE	Δ ARE
Unconditioned baseline	0.161	4.39°	—	—
Naïve self-refine (iter 5)	0.195	3.60°	+21%	−18%
Hybrid MC (iter 2)	0.160	3.13°	−1%	−29%

The Sim(3) scale ambiguity in Pi3X outputs means that conditioned translations accumulate bias across iterations. The hybrid approach avoids this by anchoring translations to the unconditioned source.

4.2 Noise Robustness of Pose Conditioning

We test Pi3X’s conditioning with ground truth poses corrupted by varying levels of Gaussian noise:

Table 3: Effect of conditioning pose quality on Pi3X output (after full pipeline). Translation noise is isotropic Gaussian in the scene coordinate system.

Rot. noise ($^\circ$)	Trans. noise	ATE↓	ARE↓
0 (perfect GT)	0	0.097	1.22°
1	0.05	0.087	1.30°
2	0.10	0.080	1.60°
5	0.20	0.077	3.34°
10	0.50	0.368	180°
20	1.00	0.951	180°

Key observations:

- Conditioning tolerates up to $\sim 5^\circ$ of rotation noise
- Beyond 10° , the conditioning signal becomes misleading and breaks the model
- Our self-conditioned prior ($\sim 4.4^\circ$ error) is at the edge of the useful regime, explaining the diminishing returns beyond 2–3 iterations

4.3 Convergence Behavior

The iterated hybrid MC method converges within 2–3 iterations:

Table 4: Convergence of iterated hybrid MC refinement (4 samples per iteration).

Iteration	ATE	ARE ($^\circ$)	RPE rot ($^\circ$)	RPE trans
0 (unconditioned)	0.161	4.39	6.26	0.375
1	0.162	3.43	4.88	0.321
2	0.160	3.13	4.58	0.309
3	0.163	3.38	4.70	0.307
4	0.160	180°	4.78	8.55

Iteration 2 is the sweet spot. Beyond iteration 3, the conditioned rotations begin to drift and can cause orientation flips (iter 4). We recommend stopping after 2–3 iterations or when ARE improvement is < 0.1 .

4.4 Sim(3) Constraint Does Not Help

We tested Sim(3)-aligning the conditioned output back to the conditioning frame before each iteration (to prevent scale drift). Results are identical to naïve iteration, confirming that translation degradation is not caused by scale drift but by systematic bias in how Pi3X uses pose conditioning:

Table 5: Sim(3)-constrained vs. naïve iteration (5 iterations).

Iter	Sim3 ATE	Sim3 ARE	Naïve ATE	Naïve ARE
0	0.152	4.69°	0.152	4.69°
1	0.160	3.50°	0.160	3.50°
2	0.180	3.31°	0.179	3.29°
5	0.197	3.61°	0.195	3.59°

5 Theoretical Interpretation

5.1 Contraction Mapping on $\text{SO}(3)$

Let $f : \text{SO}(3)^N \rightarrow \text{SO}(3)^N$ denote the mapping from conditioning rotations to Pi3X’s output rotations (after pipeline processing). If f is a contraction, i.e.,

$$d_{\text{SO}(3)}(f(R), R^*) \leq \lambda \cdot d_{\text{SO}(3)}(R, R^*) \quad (4)$$

for some $\lambda < 1$ and true rotations R^* , then by the Banach fixed-point theorem, iteration $R^{(n+1)} = f(R^{(n)})$ converges to a fixed point.

Empirical evidence. From the noise robustness experiment:

- Input error $4.4^\circ \rightarrow$ output error 3.5° ($\lambda \approx 0.80$)
- Input error $5.0^\circ \rightarrow$ output error 3.3° ($\lambda \approx 0.67$)
- Input error $2.0^\circ \rightarrow$ output error 1.6° ($\lambda \approx 0.80$)

The contraction rate is approximately $\lambda \approx 0.8$ in the $2\text{--}5^\circ$ regime, predicting convergence to a fixed point at:

$$\text{ARE}^* \approx \frac{\lambda}{1 - \lambda} \cdot \epsilon_{\text{model}} \approx 4 \cdot \epsilon_{\text{model}} \quad (5)$$

where ϵ_{model} is Pi3X’s irreducible error. With $\text{ARE}^* \approx 3.1$, this gives $\epsilon_{\text{model}} \approx 0.8$, consistent with the 1.2° achieved under perfect GT conditioning (which includes pipeline noise).

Failure for translation. Pi3X outputs poses in $\text{Sim}(3)$, not $\text{SE}(3)$. The translation component t^c is defined only relative to the per-run scale. Feeding translations back as $\text{SE}(3)$ conditioning introduces a scale mismatch that does not contract:

$$t^{(n+1)} = s^{(n)} \cdot f_t(R^{(n)}, t^{(n)}) + \text{bias} \quad (6)$$

where $s^{(n)}$ is the unknown per-run scale. This explains why ATE degrades monotonically under naïve iteration.

5.2 Denoising Score Matching Interpretation

The Monte Carlo perturbation strategy connects to denoising score matching [?]. If Pi3X was trained with noisy pose conditioning (likely, for robustness), then conditioned inference implicitly computes:

$$\mathbb{E}[R^* \mid R + \epsilon, \text{imgs}] \approx R + \sigma^2 \nabla_R \log p(R \mid \text{imgs}) \quad (7)$$

By running S perturbations $\{R + \epsilon_s\}_{s=1}^S$ and averaging the outputs:

$$\bar{R} = \text{FréchetMean}(\{f(R + \epsilon_s)\}_{s=1}^S) \quad (8)$$

we approximate the posterior mean $\mathbb{E}[R^* \mid \text{imgs}]$ via Monte Carlo integration. This is a form of Langevin Monte Carlo on the rotation manifold, where each conditioned inference step provides a sample from the posterior.

5.3 Connection to Diffusion Models

The iterative refinement process resembles a single-step denoising diffusion:

1. Start with a noisy estimate $R^{(0)}$ (unconditioned Pi3X output)
2. Apply a learned denoiser f conditioned on the images
3. The denoiser moves the estimate toward the clean manifold

Unlike standard diffusion which requires many small steps from pure noise, our setting starts close to the solution ($\sim 4^\circ$ error) and requires only 2–3 denoising steps. The key difference is that our “noise schedule” is not designed but inherited from Pi3X’s prediction uncertainty.

6 Discussion

6.1 Practical Recommendations

1. **Always run the baseline pipeline** (Stage 1)—gauge fixing, outlier rejection, and pose graph optimization provide consistent improvements at negligible cost.
2. **One hybrid refinement step** (Stage 3) offers the best cost-benefit tradeoff: $2\times$ compute for $\sim 21\%$ rotation improvement.
3. **Full MC refinement** (Stage 4) is recommended when compute budget allows: $13\times$ compute for 28% rotation improvement.
4. **Never iterate naïvely**—always use hybrid fusion to protect translations from Sim(3) bias accumulation.
5. **Stop after 2–3 iterations**—monitor ARE convergence and terminate when improvement < 0.1 .

6.2 Limitations

- The method cannot break the $\sim 3^\circ$ ARE floor without external priors (the contraction rate $\lambda \approx 0.8$ means the fixed point is at $\sim 3^\circ$)
- Translation improvement is bounded by the unconditioned baseline’s accuracy
- The $13\times$ compute cost of full MC refinement may not justify the marginal gain over a single hybrid step for latency-sensitive applications
- Convergence is empirical, not guaranteed—iteration 4+ can diverge due to orientation flips
- The method assumes access to intrinsic camera parameters (available from COLMAP or manufacturer specs)

6.3 Future Directions

- **Depth conditioning:** Pi3X also accepts depth maps as input. Monocular depth priors (e.g., Depth Anything v2) could provide complementary geometric signal without requiring ground truth poses.
- **Partial conditioning:** Condition on a subset of cameras with higher confidence, letting the model propagate information to uncertain cameras.
- **Adaptive noise scheduling:** Learn the optimal perturbation magnitude per iteration rather than using a fixed $\sigma_R = 2$.
- **Learned fusion weights:** Replace the $\max(w^{\text{cond}}, w^{\text{uncond}})$ heuristic with learned per-edge reliability.
- **Breaking the 3° floor:** Incorporate sparse LiDAR points or a pre-calibrated camera pair to provide a stronger anchor for the conditioning signal.